Kassahun Aweke Arega

Montreal (QC), Canada
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PROFESSIONAL SUMMARY

- PhD researcher in hydrogeology at Polytechnique Montréal specializing in stochastic groundwater modelling and inversions, uncertainty quantification, geostatistics, and surrogate modelling. Published researcher with university teaching and leadership experience and practical consulting contributions to groundwater resource assessment and managed aquifer recharge projects supported by international organizations.

EDUCATION

May 2025 – now

- PhD in Mineral Engineering, Polytechnique Montreal (QC), Canada
- Project: Geostatistical and Surrogate Approaches for Stochastic Recharge and Groundwater Modelling in the Harricana sub-Watershed, Abitibi–Témiscamingue
- Relevant coursework: Geostatistics; Probabilistic Machine Learning; Hydrology; Project (*Spatial Reconstruction of Annual Precipitation in the Upper Awash Basin, Ethiopia: A Comparison of Graph Convolutional Networks and Ordinary Kriging*)
- CGPA: 4.0/4.0

Dec 2020 – Jul 2022

MSc. in Geological Engineering (Hydrogeology), Addis Ababa University.

- CGPA: 3.89/4.0
- Thesis: Estimation of groundwater storage change using GRACE satellite data: "Excellent MSc thesis"
- Relevant Coursework: Groundwater hydrology, GIS, geostatistics, geophysics, contaminant transport modeling

Sep 2014 – Jul 2018

BSc degree in Geology, Woldia University.

- CGPA: 3.91/4.0 ("Highest Distinction")
- Project: Impact of landslides on road infrastructure

PROFESSIONAL EXPERIENCE

Mar 2023 – Dec 2024

Head, Department of Geology, Dire Dawa University, Dire Dawa, Ethiopia

- Led academic and administrative activities within the department
- Coordinated fieldwork, research initiatives, and collaborations

July 2022 – August 2025

Lecturer, Department of Geology, Dire Dawa University, Dire Dawa, Ethiopia

- Taught Environmental Geology and Groundwater Resources Management
- Supervised fieldwork and student research projects

Sep 2019 – Sep 2020

Graduate Assistant, Department of Geology, Dire Dawa University

- Contributed to department establishment, curriculum development, and lab organization

TECHNICAL SKILLS

- Programming: Python (Numpy, Pandas, Scipy, Matplotlib, Pytorch etc.)
- GIS: QGIS, ArcGIS
- Latex
- Groundwater recharge modelling
- MODFLOW, Flopy
- Geostatistics – Variogram modelling, Geostatistical simulation (TBM, SGS), Multiple point statistics (MPS)

	• Inversion (PEST, ES-MDA, Kalman filter) • Monte-Carlo methods • Machine Learning & Deep Learning
ACCOMPLISHMENT AND AWARDS	• Google Data Analytics • Coursera Machine Learning • Spatial Data Analysis • Project Management • Outstanding Undergraduate Student (4.0/4.0)
PUBLICATIONS	• Article: Arega, K.A., Birhanu, B., Ali, S. et al. Analysis of spatio-temporal variability of groundwater storage in Ethiopia using Gravity Recovery and Climate Experiment (GRACE) data. Environ Earth Sci 83, 206 (2024). https://doi.org/10.1007/s12665-024-11508-2.
CONFERENCE AND PROFESSIONAL PRESENTATIONS	• Poster Presentation: Stochastic groundwater recharge modellingin the Harrican river sub-watershed, Quebec; IAH-CGU joint meeting, May 2026, Halifax, Canada. • Arega, K. A. The Power of Satellite Missions to Monitor Hydro-Climatic Conditions of the Earth. Seminar presentation, College of Natural and Computational Sciences, Dire Dawa University, May 2023. • Arega, K. A. Analysis of Spatiotemporal Variability of Groundwater Storage in Ethiopia Using Gravity Recovery and Climate Experiment (GRACE) Data. Oral presentation at the 5th Research and Fair Week: "Emerging Technologies for Sustainable Socio-Economic Development," Addis Ababa University, May 2024. • Arega, K. A. Groundwater Suitability and Hydrogeological Mapping for Climate-Resilient WASH Services in Ethiopia. Oral presentation at the project validation workshop organized by the Ministry of Water and Energy, Adama, Ethiopia, February 2025. Project funded by UNICEF and implemented in collaboration with British Geological Survey, Aquacon Engineering PLC, and Acacia Water.
PROJECT PARTICIPATION	• Contributed to the development and validation of interactive workshops and accompanying numerical exercises for the French open educational resource (fabriqueREL) in applied geostatistics. • Contributed to hydrogeological mapping for the Climate-Resilient WASH project in SNNPR, Ethiopia, using remote sensing, GIS, and geophysical surveys (USAID-funded). • Participated in field investigations for the "Climate Change Impact on Groundwater in Ethiopia – Managed Aquifer Recharge (MAR) Assessment" project (BRGM). • Contributed to groundwater suitability mapping for the "Feasibility Study and Design of the Dire Dawa Managed Aquifer Recharge (MAR)" project (World Bank-funded).
MEMBERSHIP	• Member: International Association of Hydrogeologists (IAH) • Technical Member: Groundwater Relief, UK • Member: Canadian Geophysical Union (CGU), Canada • Member: International Association for Mathematical Geosciences (IAMG)
REFERENCES	• **Dany Lauzon**, Adjoint Professor, Department of CGM, Polytechnique Montreal, Montreal (QC), Canada *Relationship: Professor; PhD research main supervisor* Email: dany.lauzon@polymtl.ca • **Nicolas Benoit**, Research Scientist, Natural Resource Canada (GSC), Quebec City (QC),

Canada
Email: nicolas.benoit@nrcan-rncan.gc.ca \
Relationship: PhD research co-supervisor

- **Behailu Birhanu**, Associate Professor of Hydrogeology, School of Earth Sciences, Addis Ababa University, Addis Ababa, Ethiopia
  Relationship: Professor; MSc thesis Principal advisor
  Email: behailu.birhanu@aau.edu.et
- **Zenaw Tessema,** Senior Hydrogeologist and founder of Aquacon Engineering
  Relationship: worked together on R and D project
  Email: zenawt2000@yahoo.com
- **Binyam Tesfaw**, Associate Professor of Remote Sensing and Geoinformatics, School of Earth Sciences, Addis Ababa University, Addis Ababa, Ethiopia
  Relationship: Professor; MSc thesis examiner
- Email: binyam.tesfaw@aau.edu.et